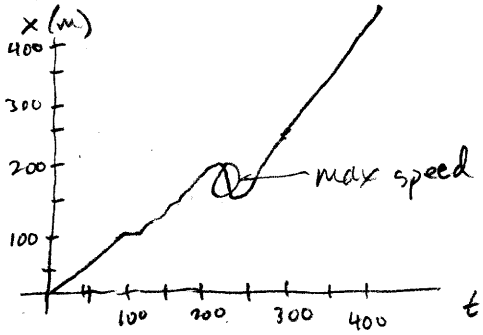
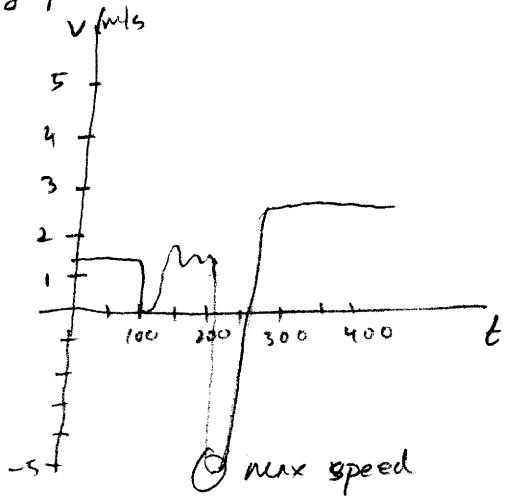
FS1 Wilkinson HW and Quiz solutions,

HW1. 1. Block length 100 m, walking speed 1.25 m/s, running 5 m/s, jog 2.5 m/s

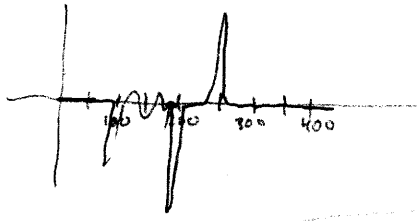
a, b





c. $\frac{400\,m}{400\,s} = 1\ m/s$

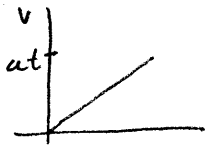
d.



short peaks of acceleration when changes direction

2 a. $a = \frac{\Delta v}{\Delta t}$ so $v = at$ if starts from zero

b.

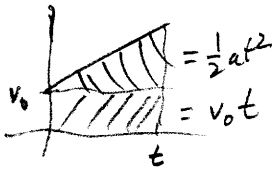


slope a

c. $\Delta x = v\Delta t$, split into small Δt, add together

d. $\frac{1}{2}(\text{base})(\text{height}) = \frac{1}{2}\cdot t\cdot(at) = \frac{1}{2}at^2$

e.



f. $\frac{dv}{dt} = a \Rightarrow v = at + C_1$, $v(0) = v_0$ so $C_1 = v_0$

$\frac{dx}{dt} = v = at + v_0 \Rightarrow x = \frac{1}{2}at^2 + v_0 t + C_2$, $x(0) = 0$ so $C_2 = 0$.

g. Use $x = v_0 t + \frac{1}{2}at^2$:

position: VW $x = 12(4) + \frac{1}{2}\cdot 4\cdot 4^2 = 48 + 32 = 80\,m$ VW ahead

Tesla $x = 0 + \frac{1}{2}\cdot 9\cdot 4^2 = 72\,m$

speed: VW $v = v_0 + at = 12 + 4\cdot 4 = 28\ m/s \approx 63$ mph

Tesla $v = 0 + 4\cdot 9 = 36\ m/s \approx 81$ mph both speeding

3 a. Top is at $t \approx 1.4$ from x plot; at this time $v = 0$

b. A: zero, constant
B: up, changing a bit
C: down, constant
D: up, changing a bit
E: zero, constant

c. A: nothing, holding ball
B: throwing
C: free movement up and down
D: catching
E: holding ball

d. $a = \frac{\Delta v}{\Delta t} \approx \frac{4}{.5} = 10\ m/s^2$

4a. $|\vec{v}|$ max, $a = 0$

b. pos. min or max, $|\vec{a}|$ max

c. Ball at top of arc

d. $\theta'' = \frac{\pi}{6}(-\omega^2 \cos\omega t) = -\frac{g}{\ell}\theta$

if $\omega = \sqrt{g/\ell}$. $\theta(0) = \pi/6$

e. Period $\omega T = 2\pi \Rightarrow T = 2\pi\sqrt{\ell/g}$

HW 1 4f. For $\theta_0 = \pi/6$. $k = \sin \pi/12 = .259$, $K(.259) = 1.598$, $T = 6.392\sqrt{\ell/g}$ actual.

Approx $T = 2\pi\sqrt{\ell/g} = 6.283\sqrt{\ell/g}$. % error $= 1.75$

For $\theta_0 = 8$ degrees, $k = .0697$, $K = 1.5727$, $T = 6.291\sqrt{\ell/g}$, % error $= 0.124$

Quiz 1 solutions already distributed

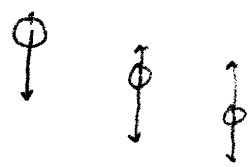
HW 2. 1. $v = at = 9.8(3) = 27.6$ m/s

2. Free fall $ma = mg$, $a = g$

3. $x = \frac{1}{2}at^2 \Rightarrow t = \sqrt{\frac{6000}{9.8}} = 24.74$ s

so $v = at = 242$ m/s $= 546$ mi/hr

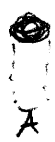
4. slower than bullet, faster than hail

5. 

air resistance increases with speed, so net F decreases and $a = \frac{F_{net}}{m}$

6. $\vec{a}$ is still down, just getting smaller, so speed is increasing but at a slower and slower rate. The speed is faster than initially, but the a is less.

7. v is constant, so $a = 0$ and the forces balance. $F_{air} = mg$.

8. 

$v\Delta t$ volume $= v\Delta t A = v\Delta t \cdot \pi R^2$

mass = vol · density $= \rho v \Delta t \pi R^2$

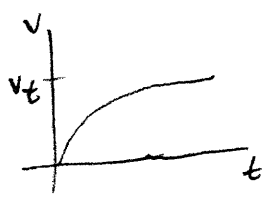
9. $mg = C\rho_{air}\pi R^2 v^2 \Rightarrow v_t = \sqrt{\frac{mg}{C\rho_{air}\pi R^2}}$

10. using $m = 3.3\times10^{-5}$ kg and $R = 2\times10^{-3}$ m (N/St).

I get $v = 6.5$ m/s

11. Only m changes to 13m in #9,

so $v_{t,Hg} = \sqrt{13}\, v_{t,water}$

12. 

13. Newton: $m\frac{dv}{dt} = mg - C\rho\pi R^2 v^2$

$= mg - mg\left(\frac{v^2}{v_t^2}\right) = \frac{mg}{v_t^2}(v_t^2 - v^2)$

so $\frac{dv}{v_t^2 - v^2} = \frac{g}{v_t^2}dt$

Integrate:

LHS: $\int_0^v \frac{dv}{v_t^2 - v^2} = \frac{1}{2v_t}\int_0^v \left(\frac{1}{v_t+v} + \frac{1}{v_t - v}\right)dv$

$= \frac{1}{2v_t}\left[\ln(v_t+v) - \ln(v_t - v)\right] = \frac{1}{2v_t}\ln\frac{v_t+v}{v_t-v}$

RHS $= \frac{gt}{v_t^2}$

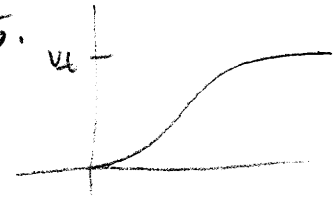
So $\ln\frac{v_t+v}{v_t-v} = \frac{2gt}{v_t} \Rightarrow \frac{v_t+v}{v_t-v} = e^{2gt/v_t}$

$\Rightarrow v = \frac{e^{2gt/v_t} - 1}{e^{2gt/v_t}+1}v_t$ (algebra).

14. Small t: $e^{2gt/v_t} \approx 1 + \frac{2gt}{v_t}$

so $v = \frac{1 + \frac{2gt}{v_t} - 1}{1 + \frac{2gt}{v_t} + 1}v_t \approx \frac{2gt}{2v_t}v_t = gt$

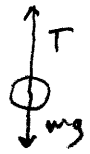
Large t: exponents dominate, $v \approx v_t$.

15. 

v_t/g for me was ≈ 0.66 sec. fast!

HW 3. 1 a. $m = 70$ kg, $T = mg = 686$ N

b. $v \approx 4$ m/s, $r \approx 3$ m, $\frac{v^2}{r} = 5.33$ m/s^2

c. 

$\frac{mv^2}{r} = T - mg$, so $T = \frac{mv^2}{r} + mg$

$= 686 + 373 = 1059$ N

54% increase with my numbers

d. - speed increases $\frac{v^2}{r}$, makes it harder to hang on
- length is tricky. If his speed is the same on a longer rope, $\frac{v^2}{r}$ is less and it's easier. But if he starts from higher, he could be moving faster. Note $v = \omega r$, so his acceleration $= \omega^2 r$; if same ω, harder to hang on with longer rope.

2 a. $\omega = \frac{2\pi}{365 \cdot 24 \cdot 60 \cdot 60} = 2 \times 10^{-7}$ rad/sec, $v = \omega r = 29.8$ km/s

b. $\frac{mv^2}{r} = \frac{GMm}{r^2} \Rightarrow v^2 = \frac{GM}{r}$, $v = \sqrt{\frac{GM}{r}}$ for circular

$G = \frac{v^2 r}{M} = 6.699 \times 10^{-11}$; actual 6.674×10^{-11}

c. $v = \sqrt{\frac{GM}{30.1 r}}$ so $v_{Nep} = \frac{v_E}{\sqrt{30.1}} = 0.182\, v_E$

year $= \frac{2\pi r_{Nep}}{v_{Nep}} = \frac{2\pi \cdot 30.1\, r_E}{.182\, v_E} = \frac{30.1}{.182}$ year$_E$

$= 165.1$ Earth year

3.

Stays on if $F_N > 0$; $F_N + mg = \frac{mv^2}{r}$.

Limit is $F_N = 0 \Rightarrow mg = \frac{mv^2}{r}$, $v = \sqrt{gr}$.

Quiz 2. 1a. $\frac{v^2}{r} = \frac{2^2}{1} = 4$ m/s^2

b.

$\frac{mv^2}{r} = T - mg$, $T = mg + \frac{mv^2}{r}$

$= 10(0.1) + 0.1(4)$

$= 1.4$ N

c. less: speed is less, so $\frac{mv^2}{r}$ decreases.

also, gravity is at an angle with T, so less T required to balance gravity.

Quiz 2 # 2.

Circular orbit $v = \sqrt{\frac{GM}{r}}$.

If r doubles, $v \Rightarrow \frac{v}{\sqrt{2}} = .70\, v_0$.

It must move at this speed, or the orbit would not be circular.

Bonus: the graph shows only the horizontal motion. At the arc bottom, centripetal a is vertically up and does not affect horizontal speed, only vertical. The graph doesn't show this.

HW 4. 1a

	U	KE	TE
6	60	0	60
4	40	20	60
2	20	40	60
0	0	60	60

b. $KE(2) = 40 = \frac{1}{2}mv^2$

$v = \sqrt{80} = 8.94$ m/s

c. U not affected
KE, TE lower.
Energy going to KE of air molecules, so heat.

2 a. Density $= \frac{\text{mass}}{\text{volume}}$! Oops!

$m = \rho V = 3(\frac{4}{3}\pi \cdot 100^3) = 12.6 \times 10^6$

b.

	U	KE	TE
100	-7.9×10^{12}	6×10^6	-7.9×10^{12}
50	-1.58×10^{13}	7.9×10^{12}	-7.9×10^{12}
10	-7.9×10^{13}	7.1×10^{13}	-7.9×10^{12}
1	-7.9×10^{14}	7.8×10^{14}	-7.9×10^{12}

c. 12 times higher

d. Initial KE is puny; hardly matters. But distance matters.

The time does depend on initial v; it is not reflected in an energy calculation.

HW 4 #3.

$$TE = \frac{-GMm}{r} + \frac{1}{2}mv^2 = -5.29\times10^{33} + 2.74\times10^{33}$$
$$= -2.55\times10^{33}$$

At ap: $-\frac{GMm}{r} = TE - KE = -2.55\times10^{33} - 2.56\times10^{33}$
$$= -5.11\times10^{33}$$

So $r = 1.55\times10^{11} = 1.035$ AU

rounding error...

4. $\Delta KE = m\frac{\Delta v}{\Delta t}\Delta x = \frac{m\Delta v}{\Delta t}\left(v_0\Delta t + \frac{1}{2}\frac{\Delta v}{\Delta t}\Delta t^2\right)$

Here, $\Delta v = v - v_0$, so

$$\Delta KE = m\left[(v-v_0)(v_0) + \frac{(v-v_0)^2}{2}\right] = m\left(vv_0 - v_0^2 + \frac{v^2 - 2vv_0 + v_0^2}{2}\right)$$
$$= m\left(\frac{v^2}{2} - \frac{v_0^2}{2}\right) = \frac{mv^2}{2} - \frac{mv_0^2}{2} \checkmark$$

5. $v_{escape} = \sqrt{\frac{2GM}{r}} = 11.2$ km/s

a. Tatoonie $M \to 2M$ $v_{esc} = \sqrt{2}(11.2) = 15.8$ km/s

b. Arrakis $r \to \frac{3}{2}r$ $v_{esc} = \frac{11.2}{\sqrt{3/2}} = 9.14$ km/s

c. Mars $M \to 0.107$, $R \to 0.532$ $v_{esc} = 11.2\sqrt{\frac{.107}{.532}} = 5.02$ km/s

smaller, mass change dominates

6 a. $\Delta KE = F\,dr$ by definition
$$= \frac{GMm}{r^2}dr$$

b. $\int_{r_0}^{r_1} \frac{GMm}{r^2}dr = \left[-\frac{GMm}{r}\right]_{r_0}^{r_1} = -\frac{GMm}{r_1} + \frac{GMm}{r_0}$

c. $r_0 \to \infty$, $U = -\frac{GMm}{r} + 0$.

d. $v = \frac{dr}{dt} = \frac{dr_m}{dt} - \frac{dr_M}{dt}$

$$\frac{dv}{dt} = \frac{d^2r_m}{dt} - \frac{d^2r_M}{dt} = \frac{F_m}{m} - \frac{F_M}{M}.$$
$$= F_m\left(\frac{1}{m} + \frac{1}{M}\right) = F_m\left(\frac{M+m}{Mm}\right) = \frac{F_m}{\mu}.$$

Sun/Earth $\mu = 5.999982\times10^{24}$.

Very close to M

7. $\frac{d}{dt}\left(\frac{m}{2}\vec{v}\cdot\vec{v}\right) = \frac{m}{2}\left(\frac{d\vec{v}}{dt}\cdot\vec{v} + \vec{v}\cdot\frac{d\vec{v}}{dt}\right) = m\frac{d\vec{v}}{dt}\cdot\vec{v}$. $\Rightarrow$ Int is $\frac{m}{2}\left(\vec{v}_1\cdot\vec{v}_1 - \vec{v}_0\cdot\vec{v}_0\right) = \frac{1}{2}mv_1^2 - \frac{1}{2}mv_0^2$.

Quiz 3. 1a. (i) at A, $KE = \frac{1}{2}mv^2$

(ii) at B, $TE = U + KE$ conserved

(iii) at B, $U = -\frac{GMm}{r}$

b. TE is conserved, so calculate U+KE at A, then for B find $U_B = TE - KE_B$ and use $U = -\frac{GMm}{r}$ to find r.

2a. $v_{escape} = \sqrt{\frac{2GM}{r}}$; r is higher, so v_{esc} lo

b. Station $r = \frac{3}{2}R$; $v_{esc} = \frac{11.2}{\sqrt{3/2}} = 9.14$ km/

HW 5 part 1.

1.a. $\langle KE\rangle = \frac{3}{2}kT = 6.1\times10^{-21}$ J for 295 K

b. $\langle KE\rangle = \frac{1}{2}m\langle v^2\rangle$. $m = 28\cdot1.67\times10^{-27}$

$\Rightarrow \langle v^2\rangle = 2.6\times10^5$ $v_{rms} = 511$ m/s

c. O_2 mass higher but $\langle KE\rangle$ same, so slow

d. $v\Delta t$ $V = v\Delta t\,\pi R^2$

$\Delta t = \frac{1}{nv_{rms}\pi R^2} = 2.82\times10^{-10}$ s

2a. N is decreasing. Air rushes out of the hole at the bottom

b. less mass in balloon → lower density ⇒ floats since weighs less than displaced air.

3a. $(70-32)\frac{5}{9} + 273 = 294$

b. If V const, $\frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow T_2 = \frac{P_2}{P_1}T_1 = 728$ K
$= 851°F$

c. $T_2 = \frac{P_2V_2}{Nk} = \frac{P_2}{Nk}\left(\frac{P_1}{P_2}\right)^{1/\gamma}\cdot V_1 = \frac{P_1V_1}{Nk}\cdot\frac{P_2^{1-1/\gamma}}{P_1^{1-1/\gamma}}$

$= T_1\left(\frac{P_2}{P_1}\right)^{(1-1/\gamma)}$ since $T_1 = \frac{P_1V_1}{Nk}$.

d. $T_2 = 294\left(\frac{260}{105}\right)^{.107} = 324$ K
$= 123°F$

e. Not adiabatic! There is heat flow. But density stays almost the same, so

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}.$$

HW 5 part 1

#4. a. $U = \frac{Nk}{\gamma-1}T$ so $dU = \frac{Nk}{\gamma-1}dT$.

But $T = \frac{PV}{Nk}$ so $dT = \frac{1}{Nk}(PdV + VdP)$

Thus $dU = \frac{PdV + VdP}{\gamma-1}$.

Also $Q=0$, so $dU = -PdV$ by First law

(we need negative sign because $U\uparrow$ for $V\downarrow$

$\Rightarrow \frac{PdV+VdP}{\gamma-1} = -PdV \Rightarrow VdP = -\gamma PdV \Rightarrow -\gamma\frac{dV}{V} = \frac{dP}{P}$

b. $\int\frac{dP}{P} = -\gamma\int\frac{dV}{V} \Rightarrow \ln P = -\gamma \ln V + C \Rightarrow P = V^{-\gamma}e^{C}$

$\Rightarrow PV^{\gamma} = \text{constant}$

c. $\gamma = \frac{n+5}{n+3} \Rightarrow \gamma n + 3\gamma = n+5$

$n = \frac{5-3\gamma}{\gamma-1} \approx 14$ (plus 3 translational KE modes)

HW 5 Part 2

2a. Non-isobaric, P changes when V does. Need calculus.

b. $18g = .018$ kg

$V_{water} = \frac{.018}{958} = 1.88\times10^{-5}\ m^3$

c. $V = \frac{nRT}{P} = \frac{1(8.314)(373)}{101300} = .0306\ m^3$

d. $P\Delta V = 101300(.0306 - 1.88\times10^{-5}) = 3099$ J/mole

e. Most of the energy goes to overcoming the intermolecular attractive forces in the liquid.

3a. ab: Pressure increasing at constant V; must be some heat going in, or an explosion

ca: V decreasing while P constant; piston could be compressing while chamber open to outside pressure to keep it constant.

3b. ab: $W=0$ since $\Delta V = 0$

$\Delta U > 0$ since $T\uparrow$ from $P\uparrow$

Thus $Q>0$ by First Law

bc: $Q=0$ since adiabatic

$W<0$ since expanding

Thus $\Delta U<0$ by First Law. Cooling

ca: $W>0$ since being compressed ($\Delta V\downarrow$)

But $\Delta U<0$ since $T\downarrow$ ($P_aV_a < P_cV_c$)

Thus $Q<0$ by First Law

c. The area enclosed

d. Look at ΔU. ab $U\uparrow$, bc $U\downarrow$, ca $U\downarrow$

Thus $T_b > T_c > T_a$.

4. a. 2-3, expansion (V increasing)

$Q=0$ so $\Delta U<0$ since $W<0$. Colder

b. Must be T_H since gas colder at 3.

$T = \frac{PV}{nR} = 481$ K

c. If T_c close to T_H, the gas is losing very little energy when it expands, so can't be doing much work.

e(!). $\left.\begin{array}{l}600C = 873K\\ 150C = 423K\end{array}\right\}\eta = \frac{873-423}{873} = 51.5\%$

6. a. Work is only in 3→4 and 1→2. These are isotherms, so $\Delta U = 0$ and T constant, $P = \frac{nRT}{V}$

$W_{34} = +\int_{V_2}^{V_4}\frac{nRT_H}{V}dV = nRT_H\ln\left(\frac{V_4}{V_2}\right)$, + meaning work by gas on piston

$W_{12} = -\int_{V_4}^{V_2}\frac{nRT_C}{V} = -nRT_C\ln\left(\frac{V_4}{V_2}\right)$, − meaning work on gas by piston

$\Rightarrow \frac{W_{tot}}{n} = R(T_H - T_C)\ln\frac{V_4}{V_2}$.

b. $Q_{23} = \Delta U$ since $\Delta V = 0$.

$= nC_v\Delta t$ so $\frac{Q_{23}}{n} = C_v(T_H - T_C)$.

~~6c. $Q_{34} = W_{34}$ since $\Delta U = 0$ on isotherm~~

~~$\Rightarrow \frac{Q_{34}}{n} = RT_H\ln\left(\frac{V_4}{V_2}\right)$~~

~~$\Rightarrow \eta = \frac{W}{Q_{in}} = \frac{R(T_H-T_C)\ln\frac{V_4}{V_2}}{C_v(T_H-T_C) + RT_H\ln\left(\frac{V_4}{V_2}\right)} = \frac{T_H - T_C}{T_H + \frac{C_v(T_H-T_C)}{R\ln(V_4/V_2)}}$~~

6c. $Q_{34} = W_{34}$ since $\Delta U = 0$ on isotherm.

So $\frac{Q_{34}}{n} = RT_H \ln\left(\frac{V_4}{V_2}\right)$.

Finally, $\eta = \frac{W_{net}}{Q_{in}} = \frac{R\ln\left(\frac{V_4}{V_2}\right)\cdot(T_H - T_C)}{RT_H \ln\left(\frac{V_4}{V_2}\right) + C_V(T_H - T_C)}$

$$= \frac{T_H - T_C}{T_H + \frac{C_V(T_H - T_C)}{R\ln(V_4/V_2)}}.$$

For $\frac{V_4}{V_2} = 10$ and $C_V = \frac{5}{2}R$, $\eta = \frac{T_H - T_C}{T_H + \frac{5}{2\ln 10}(T_H - T_C)} = \frac{T_H - T_C}{T_H + 1.08(T_H - T_C)}$

With $T_H = 400$ and $T_C = 300$, $\eta = \frac{100}{400 + 108} = 19.7\%$

Carnot $\eta = \frac{T_H - T_C}{T_H} = \frac{100}{400} = 25\%$

Closer for higher compression ratio.